Fig 5e v3 AF rewiring mechanism cartoon (asymmetric, NCP_v3)

All 4 *** Priority families are LA-driven (4 / 4)

Source: atlas ChamberDriver column (Fig 2D pooled g + per-cohort delta_LA / delta_RA)

RA-axis (left tier)

no high-priority families

RA-driven families (all * PriorityClass):

- **F7 TCA cycle**
  carrier: Mix ρ = -0.30 (Single-axis)
- **F8 Complex I biogenes**
  carrier: Endo ρ = +0.25 (Single-axis)
- **F6 Mito protein impor**
  carrier: Mix ρ = +0.11 (Silent)

Interpretation:

AF chamber rewiring at the highest-priority tier is concentrated on the LA arm; the RA arm shows only low-priority single-axis trends (mostly metabolic OXPHOS-related).

LA-axis (right tier) • 6 priority families (4 LA-driven + 2 Bilateral)

carrier: Fibro

F15 NTRK2 / TrkB signalli…
L3 dur ρ = -0.39 (Weak-trend)
L2 fidelity: Relay
L1 replication: high
cross-axis: Triple-positive
***
LA-driven

carrier: Endo

F11 BMP signalling
L3 dur ρ = -0.20 (Stable)
L2 fidelity: Below-detection
L1 replication: high
cross-axis: Single-axis
***
LA-driven

carrier: Fibro

F9 HS-GAG metabolism
L3 dur ρ = -0.42 (Strong-trend)
L2 fidelity: Below-detection
L1 replication: high
cross-axis: Double-positive
***
LA-driven

carrier: Fibro

F10 TGF-beta signalling (H
L3 dur ρ = -0.22 (Weak-trend)
L2 fidelity: Below-detection
L1 replication: high
cross-axis: Double-positive
***
LA-driven

carrier: aCM

F2 Mito Ca2+ transport
L3 dur ρ = -0.56 (Strong-trend)
L2 fidelity: Below-detection
L1 replication: moderate
cross-axis: Single-axis
*
Bilateral

carrier: Adipo

F4 BCAA catabolism
L3 dur ρ = -0.03 (Stable)
L2 fidelity: Below-detection
L1 replication: moderate
cross-axis: Silent
*
Bilateral

Chip encoding: background = PriorityClass (***/**/*); border colour = cross-axis class (5C); top strip = HCA LA carrier (cell-type, Fig 1F); middle = L3 dur ρ + chronicity / L2 fidelity / L1 replication; bottom = PriorityStars + Chamber

Honest framing: Asymmetry is the message: no atlas Top-priority family is RA-driven. Bilateral chips (F2, F4) are placed in the right tier because Bilateral includes the LA arm.